Eduzen — Global Enrichment Programme

Chemistry | IGCSE | Grade 9 | Topic: Relative Masses of Atoms and Molecules | **80 Marks**

Name: ______________________________ Date: _____________________ Grade: **9**

General Instructions

1. A_r values: H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, Al = 27, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Cu = 63.5, Zn = 65
2. Show all working for numericals. Include units where needed. 3. $L = 6.02 \times 10^{23}$ mol^{-1}

Section A: Objective Questions (1 × 20 = 20 Marks)

Circle the correct option.

1. The relative atomic mass of an element is measured compared to:

a) ^{1}H atom b) ^{12}C atom

c) ^{1}■O atom d) ^{1}■N atom

Ans: [ ]

2. What is the M_r of H_2SO_4?

a) 49 b) 98

c) 100 d) 96

Ans: [ ]

3. 1 mole of CO_2 contains:

a) 6.02×10^{23} molecules b) 44 g

c) 22.4 dm^3 at rtp d) All of these

Ans: [ ]

4. Which has the highest mass?

a) 1 mol of H_2O b) 1 mol of CO_2

c) 1 mol of NH_3 d) 1 mol of CH_4

Ans: [ ]

5. A_r has no units because:

a) It is very small b) It is a ratio

c) It is measured in grams d) It is an average

Ans: [ ]

6. Number of atoms in 0.5 mol of oxygen gas O_2:

a) 3.01×10^{23} b) 6.02×10^{23}

c) 1.204×10^{24} d) 1.505×10^{23}

Ans: [ ]

7. The M_r of calcium carbonate $CaCO_3$ is:

a) 50 b) 68

c) 100 d) 116

Ans: [ ]

8. 2 moles of NH_3 weigh:

a) 17 g b) 34 g

c) 8.5 g d) 28 g

Ans: [ ]

9. Which statement is true for isotopes?

a) Same A_r
b) Same no. of neutrons
c) Different A_r
d) Same chemical properties and mass

Ans: [ ]

10. Mass of 3.01×10^{23} molecules of H_2O is:

a) 9 g
b) 18 g
c) 36 g
d) 4.5 g

Ans: [ ]

11. The formula mass of $Mg(OH)_2$ is:

a) 41
b) 58
c) 42
d) 59

Ans: [ ]

12. 0.25 moles of Cl_2 gas contains how many molecules?

a) 1.505×10^{23}
b) 6.02×10^{23}
c) 3.01×10^{23}
d) 2.408×10^{24}

Ans: [ ]

13. A_r of chlorine is 35.5 because:

a) Cl has 35.5 protons
b) Average of isotopes ^{3}■Cl and ^{3}■Cl
c) Cl is unstable
d) It's rounded off

Ans: [ ]

14. Which contains the most atoms?

a) 1 g H_2
b) 1 g O_2
c) 1 g C
d) 1 g N_2

Ans: [ ]

15. M_r of hydrated copper(II) sulfate $CuSO_4{\cdot}5H_2O$:

a) 159.5
b) 177.5
c) 249.5
d) 250

Ans: [ ]

16. 4 g of oxygen gas at rtp occupies:

a) 3.0 dm^3
b) 6.0 dm^3
c) 1.5 dm^3
d) 12.0 dm^3

Ans: [ ]

17. The mass of one atom of carbon-12 is:

a) 12 g
b) 2×10^{-23} g
c) 1.99×10^{-23} g
d) 6.02×10^{23} g

Ans: [ ]

18. Number of moles in 8.8 g of CO_2:

a) 0.1
b) 0.2
c) 0.4
d) 2

Ans: [ ]

19. Which pair has the same M_r?

a) CO_2 and NO_2
b) N_2 and CO
c) O_2 and SO_2
d) H_2O and NH_3

Ans: [ ]

20. Empirical formula mass is used when:

a) Compound is covalent
b) Compound is ionic
c) Calculating moles
d) Finding M_r

Ans: [ ]

Section B: Short Numericals (2 × 10 = 20 Marks)

Show all working clearly.

21. Calculate the M_r of glucose $C_6H_{12}O_6$.

22. Find the mass of 0.3 moles of NaOH.

23. How many moles are in 25 g of $CaCO_3$?

24. Calculate the number of molecules in 11 g of CO_2.

25. What is the mass of 1.204×10^{24} molecules of H_2O?

26. Find the A_r of element X if 0.5 mol of X weighs 16 g.

27. Calculate the formula mass of $Al_2(SO_4)_3$.

28. How many atoms are there in 4 g of helium gas?

29. Find the volume of 3.2 g of O_2 at rtp. *(Molar volume = 24 dm^3)*

30. 0.2 mol of a gas has mass 5.6 g. Find its M_r.

Section C: Descriptive Questions (3 × 5 = 15 Marks)

31. Define relative atomic mass. Why is ^{12}C used as the standard?

32. Distinguish between relative molecular mass and relative formula mass with examples.

33. Explain how A_r of chlorine is 35.5 using the concept of isotopes. Include % abundance: ^{3}■Cl = 75% and ^{3}■Cl = 25%.

34. Why do we use the mole concept? Explain using the relationship between mass, M_r, and number of particles.

35. A compound has empirical formula CH_2 and $M_r = 56$. Deduce its molecular formula.

Section D: Story-Based & Applied Questions (5 × 5 = 25 Marks)

Q36. The Mystery White Powder [5 Marks]

A forensic scientist finds 5.85 g of a white powder at a crime scene. Analysis shows it contains 0.1 mol of formula units. The powder is suspected to be NaCl.

a) Calculate the M_r of the powder from the data.

b) Calculate the M_r of NaCl using A_r values.

c) Is the powder likely to be NaCl? Explain.

Q37. Baking Soda in the Kitchen [5 Marks]

A recipe needs 8.4 g of baking soda $NaHCO_3$. Your kitchen scale is broken, but you have a mole spoon that measures 0.05 mol.

a) Calculate M_r of $NaHCO_3$.

b) How many moles are in 8.4 g?

c) How many spoonfuls do you need?

Q38. Industrial Oxygen Tanks [5 Marks]

A factory uses oxygen gas for welding. One tank contains 16 kg of O_2.

a) How many moles of O_2 are in the tank?

b) How many O_2 molecules is this?

c) What volume would this occupy at rtp?

Q39. Hydrated Salt in the Lab [5 Marks]

A student heats 25.0 g of hydrated copper(II) sulfate $CuSO_4 \cdot 5H_2O$ to remove water. After heating, 16.0 g of anhydrous $CuSO_4$ remains.

a) Calculate the mass of water lost.

b) Calculate moles of anhydrous $CuSO_4$ formed.

c) Calculate moles of water lost. Hence find x in $CuSO_4 \cdot xH_2O$.

Q40. The Carbon Footprint [5 Marks]

A car emits 132 g of CO_2 per km.

a) How many moles of CO_2 are emitted per km?

b) How many CO_2 molecules is this?

c) If a person drives 50 km daily, what mass of carbon alone is released per day?